Evaluation of the learned and optimising components

Generated by `pnpm report` at 2026-10-20T14:20:53.949Z from commit `1ae830c34b8d40add73c509b61444a08c37285f4`. Every number is recomputed on its fixed seed; the raw values are in `docs/results/`. Charts of the same runs are in `docs/charts/`.

All data is synthetic and says so: the message corpus is hand written, the flights are generated from physical rules with noise. Each component has a held out split chosen by a fixed seed before training, and the agent has no write path to the corpus, the seed or the threshold. None of these components signs, pays or changes a cap.

1. Message classifier (`ml/gate.ts`)

Logistic regression on hashed character 1 to 3 grams, 2^14 buckets, trained at module load. It adds a risk score the rule gate can only turn into a hold.

Metric	Value
Training messages	262
Held out messages	87
Accuracy	93.1%
Precision	100.0%
Recall	81.3%
F1	0.897
False alarm rate	0.0%
Hold threshold	0.8
Seed	20251020

87 of 349 messages held out by seed 20251020, never trained on; the 52 scripted attack texts are excluded from training. Precision 100.0% (95% CI 87.1 to 100.0, 26 of 26), recall 81.3% (95% CI 64.7 to 91.1, 26 of 32), false alarm 0.0% (95% CI 0.0 to 6.5, 0 of 55). On the 52 unseen scripted attack texts the classifier alone labels 27 as attacks and would hold 13 at threshold 0.8; the 52 of 52 red team result is the full pipeline (rule gate, classifier, policy engine, contract mirrors).

2. Telemetry oracle (`ml/plausibility.ts`, `fusion/kalman.ts`, `fusion/conformal.ts`)

Robust z over five features of a flight log, the GPS track smoothed by a Kalman filter with a Rauch Tung Striebel pass, the flag threshold set by split conformal calibration for a 2% false alarm target.

Metric	Value
Flights	2000 generated, 500 held out, 72 of them tampered
Detection	100.0%
False alarm	3.0%
Conformal threshold	z 2.60 for a 2.0% target, 634 calibration flights

Metric	Value
Coverage	97.0%
Raw track false alarm at matched detection	13.8%
Smoothed track false alarm at matched detection	0.0%
Seed	7331

Tamper class	Flights	Caught
inflated_mu	15	15
replayed_track	16	16
impossible_speed	16	16
battery_too_low	15	15
teleport_jump	10	10

500 of 2000 synthetic flights held out by seed 7331, 72 tampered; fit on 750, conformal calibration on the honest rows of the next 750. Detection 100.0% (95% CI 94.9 to 100.0, 72 of 72); false alarm 3.0% (95% CI 1.8 to 5.1, 13 of 428) on 428 honest flights of the 500 held out (seed 7331), full five feature model plus replay and teleport checks at the conformal threshold z 2.597. The 0.0% smoothed figure is an ablation on the two track features at a fixed z 4, not the oracle's false alarm rate.

3. Lease strategy (`economics/mdp.ts`)

A finite horizon Markov decision process over lease balance, bond, weather and offer class, solved by value iteration, compared with the greedy rule on the same seeded seasons with common random numbers.

Metric	Greedy rule	MDP policy
Mean season profit after instalments	7,471 yuan	7,540 yuan
Standard deviation	4,882 yuan	4,837 yuan
Seasons reaching title	84.0%	84.0%

46,800 states, 26 weeks, 200 seasons, seed 2026, advantage 69 yuan a season.

4. Bond stress test (`economics/stress.ts`)

Metric	Value
Seasons	2000
Default probability	9.0%
Lender loss given default, mean, no bonds	21,081 yuan
Lender loss given default, p95, no bonds	25,593 yuan
Lender loss given default, mean, with bonds	14,582 yuan
Lender loss given default, p95, with bonds	19,739 yuan
Share of the p95 loss the bonds absorb	23.0%
Expected vendor bond drawdown a season	358 yuan

Metric	Value
Expected pilot slashes a season	346 yuan
Seed	9001

5. Strategy ratchet (`ratchet.ts`)

Metric	Value
Proposals	60
Kept	5
Broke the reserve rule	10
Profit on the frozen week, baseline	246 yuan
Profit on the frozen week, best	926 yuan
Seeds	eval 4242, walk 99

6. Pilot credit score (`credit/score.ts`)

Buhlmann credibility on one fixed sequence: the fleet prior, three confirmed jobs, one held log. Score path 0.912, 0.922, 0.930, 0.936, 0.882; final tier B, instalment share 2815 bps inside the band.

7. Drift monitor (`drift/cusum.ts`)

One sided CUSUM on the classifier's risk score (target 0.1, k 0.25, h 2). alarm 6 observations after the shift, 0 false alarms in 600 benign scores, synthetic scores on seed 20251020.

8. Second price planner (`planner.ts`, `economics/pricing.ts`)

second price: the higher margin per mu flies first and pays the runner up's bid in priority, so an inflated fee cannot buy a slot at a profit.